Development of a Portable Capacitance Metre

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***Abstract*— This report details the development of a portable, low-cost capacitance metre using a differential charge/discharge circuit for accurate, stray-immune measurements. The aim is to deliver a high-resolution, ±1% accurate device tailored to the needs of students, field engineers, and small manufacturers seeking reliable and affordable measurement tools. Existing market options often lack calibration, are expensive, or compromise portability. In response, the proposed system integrates a microcontroller-based RC timing method, embedded software calibration, and a simple LCD interface within a compact, battery-powered design. The implementation prioritises cost-efficiency, robustness, and ease of use, offering improved performance over comparable DIY and commercial alternatives. Careful component selection and PCB layout minimise parasitic effects, making the device well-suited for both lab and field applications.**

I. Introduction

Accurate, low-cost capacitance measurement is a core need in electronics diagnostics, materials sensing, and component testing. Many users—particularly students, field engineers, educators, and small-scale manufacturers—require a compact and reliable tool for measuring small capacitance values with minimal setup [1]. While commercial LCR metres offer excellent performance, they are often expensive, bulky, or lack customisability [2]. DIY capacitance metres, on the other hand, frequently suffer from noise, limited resolution, or lack calibration functionality [3], [4]. This project aims to address that gap by developing a portable, battery-powered capacitance metre based on a differential charge/discharge circuit with microcontroller-based timing and embedded calibration [1], [5]. Designed to be produced for under £75, this system targets educational, maker, and diagnostic markets where cost, simplicity, and accuracy are equally valued.

II. Review of Existing technologies

The foundational work in capacitance measuring circuits introduced the differential charge/discharge circuit, which has become a benchmark for stray-immune capacitance measurement systems [1]. The design, as shown in Figure 1, uses a combination of op-amps and synchronous switching to cancel out stray capacitances (Cs1 and Cs2), making it suitable for portable systems where environmental noise may affect measurement stability.

One widely used alternative to the charge/discharge approach is the AC-based capacitance measuring circuit, which applies a sinusoidal voltage to the unknown capacitor and use a phase-sensitive detection method to extract the capacitive component from the total impedance [1], [6].

TABLE I. Comparison between Charge/Discharge and AC-based Measurement

Method	Advantages	Disadvantages
Charge/Discharge	Simple hardware, cost-effective	noisy, calibration and temperature compensation
AC-Based Measurement	High accuracy, noise immunity	Complex, requires stable oscillator and phase detection logic

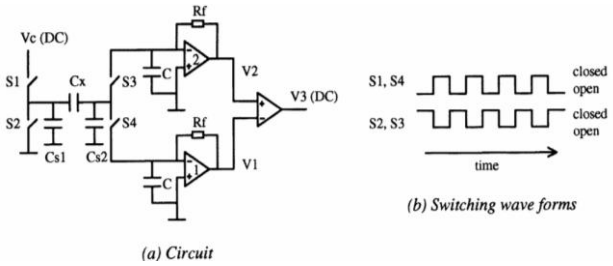


Figure 1. Charge/Discharge Circuit with Switching Wave Forms [1].

A relevant example is described in US Patent US8446157B2 from ABB Technology AG, which employs an AC-based method using a known excitation frequency and measures both current and voltage to derive capacitance via impedance calculations [7]. The system benefits from high accuracy and good noise rejection but is significantly more complex in terms of circuitry and calibration [8]. Table I outlines the key distinctions between these two methods.

Recent developments using the charge/discharge approach have adapted this principle to microcontroller platforms. Including an implementation of a transient-based charge/discharge measurement technique on an Arduino-compatible microcontroller, providing a cost-effective and compact solution with software-controlled calibration and timing [9].

Al-Hamry et al. developed a low-cost system using a relaxation oscillator and active bridge circuit, achieving wireless data transfer and user interface integration for IoT applications [3]. This approach demonstrates the feasibility of embedding capacitance metres in portable, connected devices.

The patent EP3654028A1 from D. Bez describes a capacitive humidity sensor circuit using microcontroller-timed charge/discharge cycles [10]. The design's minimal analogue components and embedded calibration make it directly relevant to the goals of this report.

These developments show a convergence towards microcontroller-based systems for portable capacitance measurement. However, most existing solutions lack the combination of stray immunity, high resolution, and calibration flexibility offered by integrating the original differential architecture with modern microcontroller techniques. Thus, the proposed device is unique in offering Differential charge/discharge measurement, Stray capacitance immunity, Software-based calibration routines and Compact, battery-powered operation with display interface.

III. Product Proposal

A. Overview and Working Principle

The proposed product is a handheld portable capacitance metre that employs a differential charge/discharge circuit for accurate capacitance measurement. The core working principle follows the differential configuration, which ensures stray capacitance immunity [1]. In the charging phase, a known DC voltage is applied through a high-precision resistor to the unknown capacitor. A microcontroller monitors the voltage rise and records the time taken to reach a threshold,

thereby calculating capacitance based on the RC time constant.

B. System Specifications

To guide the design and implementation of the proposed capacitance metre, a clear set of quantitative performance objectives was established. These specifications form the technical foundation of the system and influence every design decision, from component selection to circuit layout and firmware structure. The figures in Table II reflect key operational targets, including measurement range, resolution, power efficiency, and environmental robustness, all optimised for handheld use in both laboratory and field conditions [9].

TABLE II. Parameters and Target Specifications of the System

Parameter	Specification
Measurement Range	10 pF – 1000 nF
Resolution	1 pF
Accuracy	±1% (after calibration)
Power Supply	3.7V Li-ion / USB 5V
Power Consumption	<50 mW
Dimensions	~110mm x 70mm x 25mm
Weight	<200g
Operating Temperature	0°C to 50°C
Response Time	<100 ms per reading
Interface	16x2 LCD screen

C. Component Selection and Design Details

To minimise the influence of stray capacitance, the layout will feature short, shielded signal paths and may incorporate guard traces around high-impedance nodes [1]. Precision metal film resistors with low temperature coefficients will be used to reduce drift, and the capacitor under test will be connected via secure, low-parasitic terminals. Additionally, the measurement circuit is configured such that the RC time constant produces timing intervals well within the resolution capability of the microcontroller's 16-bit timer, ensuring accuracy even at low capacitance values [5]. The key components are:

- Microcontroller: ATmega328P (Arduino Nano) – £2.50
- Precision Resistor: 10 kΩ ±0.1% metal film – £0.10
- MOSFET Discharge Switch: 2N7000 – £0.10
- LCD Display: HD44780-compatible 16x2 – £2.50
- Battery: 3.7V 1000 mAh Li-ion – £3.00
- Charger Module: TP4056 USB Li-ion charger – £0.50
- Enclosure: ABS plastic – £5.00
- Miscellaneous: Capacitor clips, push buttons, headers – £1.00

The estimated base-level cost for these core components is set at £15 [9], [10].

D. System Architecture

The overall structure of the proposed portable capacitance metre is illustrated in Figure 2, which provides a block-level overview of the system's core functional units, signal flow, and user interface components. The design is centred around an 'ATmega328P' microcontroller, which manages timing, signal processing, user interaction, and output display.

The system is powered by a 3.7 V Li-ion battery, which is charged via a 'TP4056' USB charging module. The battery supplies regulate voltage to the microcontroller and all peripheral components. This power arrangement ensures portability and ease of recharging without the need for an external bench supply.

The 'ATmega328P' handles the measurement process through its internal timer, analogue comparator, and GPIOs. The comparator monitors the voltage across the capacitor under test (Cx), and the timer records the duration required for the voltage to reach a defined threshold, forming the basis for capacitance calculation. The EEPROM stores calibration data, allowing compensation for component tolerances and offset errors [10].

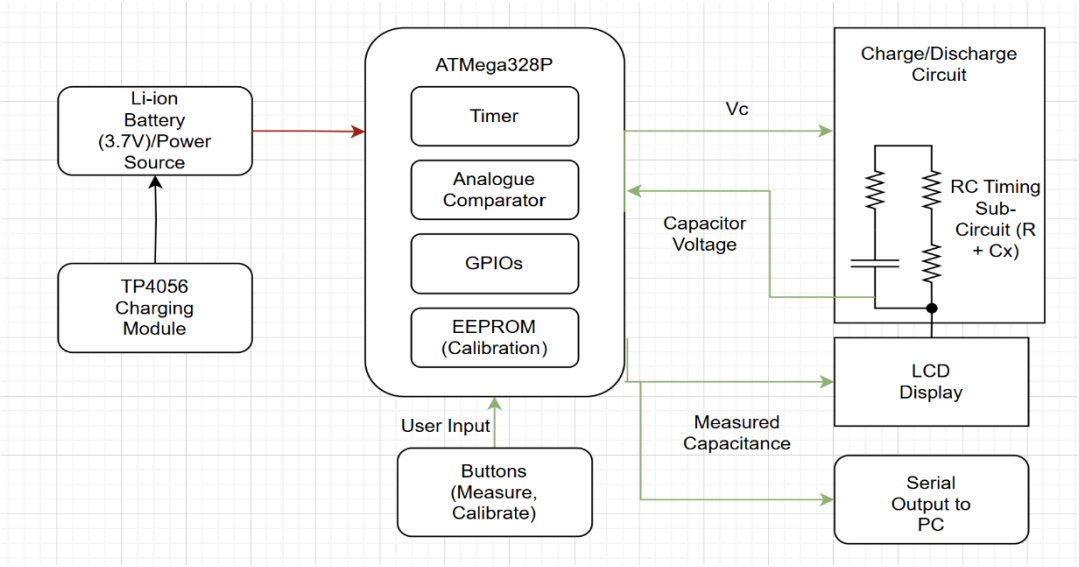


Figure 2. System Architecture Block Diagram

The measurement is performed using a dedicated charge/discharge circuit containing an RC timing sub-circuit. This sub-circuit includes a precision resistor and the unknown capacitor Cx, whose time-dependent voltage rise is tracked to determine capacitance. The microcontroller outputs a control signal (Vc) to initiate the charging cycle and receives the capacitor voltage as a feedback signal to terminate the timing process [5].

This architecture provides a low-cost, low-power, and modular solution that is easy to build and customise. The clear separation of power, processing, sensing, and output stages ensures scalability and makes the system ideal for use in academic, industrial, and field environments.

E. Software and Signal Processing

The firmware for the capacitance metre is developed in C/C++ using the Arduino IDE, optimised for execution on the 'ATmega328P' microcontroller. It controls all functional operations of the system including charging cycles, timing measurements, calibration, power management, and user interface feedback [9].

Upon powering up, the microcontroller performs hardware initialisation: it configures digital I/O for the push buttons, the LCD display, and the Vc output pin, and sets up the timer and analogue comparator. The system waits in a low-power idle mode until a button press triggers a measurement or calibration routine.

In measurement mode, the firmware sets the Vc control pin HIGH to begin charging the capacitor through a known precision resistor. Simultaneously, the timer is started. The capacitor voltage, monitored via the comparator, rises exponentially. When the voltage reaches a preset threshold, the comparator output triggers the microcontroller to stop the timer. The time duration recorded by the timer corresponds to the RC time constant of the sub-circuit [5]. Given the known resistance , the firmware calculates the unknown capacitance using the rearranged time constant formula (1):

$$C_x = \tau \div R \quad (1)$$

This result is then displayed on the LCD and optionally transmitted via the serial port to a connected PC.

The calibration mode, initiated via a long press of the designated button, prompts the user to insert known reference capacitors. The corresponding timing values are recorded and used to generate correction factors, which are stored in non-volatile EEPROM memory. These corrections compensate for internal offsets, comparator delay, and parasitic effects. The EEPROM-stored calibration data is automatically loaded at startup [9].

To improve reliability and reduce the impact of random noise or signal instability, the firmware applies a moving average filter to the measured time values. Additionally, it includes error handling routines that display warnings for disconnected components, out-of-range values, or low battery voltage.

The software is structured into clearly defined state-driven functions and interrupt service routines (ISRs) to ensure responsive and energy-efficient operation. This architecture allows for accurate and repeatable measurements under real-world operating conditions, while maintaining minimal resource consumption and fast user interaction [3].

F. User Interface and Packaging

Users interact via:

- Buttons: One for measurement, one for calibration, and one power toggle.
- LCD Display: Shows real-time capacitance values, calibration instructions, and system status.
- Port Access: USB charging and optional data connection.

The enclosure will be made of ABS plastic, handheld in size, with cut-outs for display, USB, and buttons. Components will be mounted on a custom PCB or prototyping board, using shielded wires to minimise interference.

G. Performance Estimation

Beyond the basic specifications, a deeper evaluation of system performance reveals several technical factors that influence the reliability and precision of measurements. These include time resolution limits, measurement resolution, and sources of systematic and random error.

The timing resolution of the 'ATmega328P' microcontroller is dictated by its internal 16-bit timer operating at a 16 MHz clock frequency. This results in a minimum time increment of 62.5 ns [11]. While this provides a theoretical basis for extremely fine resolution, it imposes a practical limit on how small of a capacitance change can be distinguished. For small capacitors (e.g. <50 pF), the voltage rise is extremely rapid, and the timer may only register a small number of ticks, potentially reducing accuracy due to quantisation error and comparator response lag.

Measurement resolution is also affected by parasitic capacitance from test leads, PCB traces, and even the user's hand. These parasitics add a baseline offset and reduce the dynamic range, particularly for small capacitances. While shielding and layout minimisation reduce this impact, it cannot be completely eliminated without a more complex differential architecture.

Several sources of error remain inherent to this design [12]:

- Comparator offset and hysteresis: While calibration compensates for threshold mismatch, internal hysteresis or propagation delay may still introduce nonlinearity, especially for rapid transitions.
- Microcontroller clock drift: Any deviation in clock frequency directly skews the time measurement, and therefore the calculated capacitance.
- Voltage instability: The Vc supply must remain stable during the charging cycle. A droop in Vc during measurement could affect the rate of voltage rise across the capacitor and distort timing.

The use of averaging smooths out random jitter, and calibration reduces systemic bias, but for high-precision applications requiring sub-pF resolution, additional compensation techniques or more sophisticated circuitry would be required.

Additional estimated performance metrics provide further insight into the system's capability:

- Linearity: Targeted within ±1% deviation over the range of 100 pF to 1 µF.
- Temperature Drift: <0.2% change over a 25°C to 45°C temperature range, supported by low-TC resistors.
- Long-Term Stability: Less than ±1.5% variation over 8 hours of continuous operation under stable environmental conditions..
- Power Efficiency: Estimated at 0.2 mWh per measurement, enabling over 5000 measurements per battery charge.

These metrics validate the system's practical usability, especially in field or low-power environments, while also reinforcing its performance in applications where moderate precision and repeatability are essential.

H. Construction Requirements

Equipment Needed:

- Soldering iron & hand tools.
- Oscilloscope for signal integrity.
- Precision capacitors for calibration.
- Access to multimeter, prototyping materials, optional 3D printer.

Facilities:

- Small bench area.
- USB power source.
- No special cleanroom or ESD setup required.

I. Testing Program

A structured and comprehensive testing programme is essential to ensure that the proposed capacitance metre meets the stated performance, safety, and user interface objectives. The testing process is divided into three core phases: performance testing, interface and safety testing, and environmental validation.

Performance Testing: This phase involves verifying the accuracy, resolution and repeatability of the device. A set of precision reference capacitors ranging from 10 pF to 1 µF will be used to evaluate the metre's measurement accuracy and linearity. Each capacitor will be measured multiple times to assess short-term repeatability, and results will be compared to nominal values to quantify percentage error. Temperature sensitivity will be evaluated in an ambient range of 20°C to 45°C using a temperature-controlled environment.

Safety and Interface Testing: The system's safety will be assessed under typical and boundary conditions. Battery safety will be validated by verifying charging circuit behaviour under normal and over-discharge scenarios using the TP4056 module. Button functionality will be tested for responsiveness, debouncing, and fault tolerance. The LCD will be evaluated for contrast, refresh rate, and stable performance across voltage fluctuations. Firmware-based error handling routines will also be triggered intentionally to observe failure modes and user feedback.

Environmental and Robustness Testing: Mechanical robustness will be verified by conducting minor drop and vibration tests to ensure internal components remain secured. Noise immunity will be qualitatively assessed using measurement trials in environments with nearby electromagnetic interference sources such as soldering irons and power supplies.

J. Timeline and Estimated Costs

Table III outlines the key stages and associated costs. The timeline spans 10 weeks, including planning, assembly, firmware development, calibration, and documentation. The total material cost is expected to remain under £75, consistent with comparable designs in the literature [9], [13].

Table III. Estimated Timeline and Costs

Phase	**Week Number**	**Description**	**Estimated Costs (£)**
Planning	1-2	Background research and design concept	0
Design and Component Sourcing	3-4	Schematic, layout and part acquisition	30
Construction	5	PCB/prototyping board, hardware assembly	15
Firmware Development	6-7	Software implementation	0
Calibration	8	Functional testing, calibration with standard capacitors	15
Documentation and Reporting	9	Report drafting and leaflet preparation	0
Final Testing and Validation	10	Final checks, packaging and performance verification	10
Total Estimation	**10 Weeks**		**~ 70**

The cost structure for this project reflects three tiers of expenditure. The first is the component-level cost, covering only essential hardware such as the microcontroller, resistor, LCD, battery, charging circuit, and enclosure. This totals approximately £15 and represents the minimum viable prototype. The second tier, the full prototype cost, accounts for additional materials including a PCB or protoboard, spare components, upgraded enclosures, and assembly consumables, bringing the total to around £30. Finally, a comprehensive academic project budget may approach £120 when including test equipment, high-precision reference capacitors, tools, and other development overheads.

The timeline for the project is structured over 10 weeks. The first two weeks focus on literature review and concept development. Hardware design and acquisition occur in weeks 3–4, followed by assembly and prototyping in week 5. Firmware development is scheduled for weeks 6 and 7. Calibration and validation tests are planned for week 8. Week 9 is dedicated to report writing, and final testing and adjustments are carried out in week 10.

C-Meter Pro

Precision Measurements, Minimal Footprint

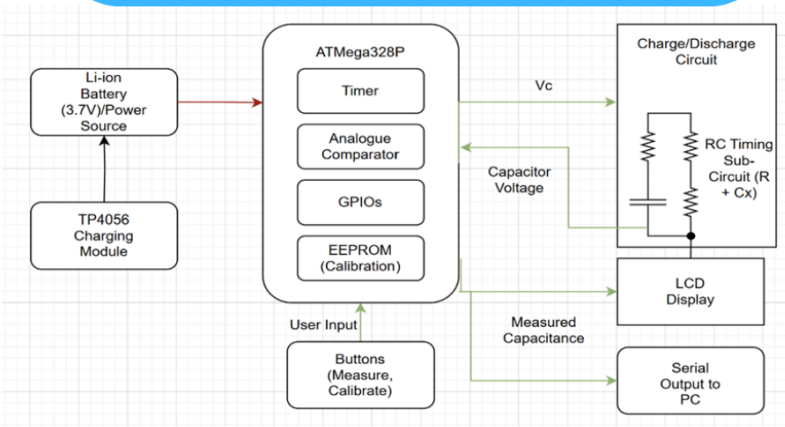
A portable capacitance meter built for the lab, the line and the field

Product Summary

The C-Meter Pro is a high accuracy, handheld capacitance metre designed for manufacturers, technicians and researchers. Delivering rapid, high pF resolution without requiring bulky benchtop instruments.

How it Works

Accurate Results, Simple Design



Key Benefits

✓ *For Manufacturers*

1) Low-cost BOM
2) Compact Design
3) Minimal Calibration after assembly

✓ *For Consumers*

1) Portable- can fit in your lab coat pocket!
2) Simple two button interface
3) LCD shows live readings

Developer: Qais Momani,University of Manchester

Technical Specs

Parameter	Specification
Measurement Range	10 pF – 1000 nF
Resolution	1 pF
Accuracy	±1% (after calibration)
Power Supply	3.7V Li-ion / USB 5V
Power Consumption	<50 mW
Dimensions	~110mm x 70mm x 25mm
Weight	<200g

Applications

On-site diagnostics

Educational labs

Quality control in manufacturing

IV. Conclusion

This report has presented the detailed design and development of a portable, low-cost capacitance metre employing a differential charge/discharge measurement circuit. By carefully reviewing existing capacitance measurement technologies, a clear market gap was identified for a device that balances accuracy, affordability, portability, and usability. The proposed system uniquely integrates a microcontroller-based RC timing method, embedded calibration routines, and robust hardware layout to deliver high-resolution (1 pF) and high-accuracy (±1%) measurements. Comprehensive performance estimations indicate the design effectively mitigates common errors such as temperature drift, comparator latency, and parasitic capacitance effects, ensuring reliable operation in practical scenarios. With an estimated production cost below £75, the device addresses the targeted market segments including educational institutions, small manufacturers, and field engineers. Future enhancements could include expanded measurement range, wireless connectivity for data logging, and further miniaturisation, making the capacitance metre even more versatile and attractive to users.

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